

SPRING 2025 (CSE423 QUIZ)

Time: 30 Minutes

Name: _____ ID: _____

Section: _____

A

Suppose you are standing on a designed floor. The floor is situated at the **XY plane**. Your eye position is **(3, 4, 6)**. There is a pattern with color— **10% red, 20% green, 70% blue**, drawn on the floor at **(1, 1, 0)**. The room has an ambient light with intensity, **$I_a = 0.2$** . Here, Ambient coefficient **$k_a = 0.4$** , Diffuse coefficient **$k_d = 0.6$** , Specular coefficient **$k_s = 0.5$** and Specular exponent **$n = 2$** . Answer the following questions based on it:

1. What is the normal vector of the plane (N)? [2]
2. Calculate the unit vector of viewing direction (V) (The vector going from pattern to Eye)? [2]
3. What color will you find in the pattern at the current setup (only ambient light)? [3]
4. Now you turned on a white light with intensity 0.7 situated at (-2, -1, 6).
 - a. Calculate the unit vector of the reflected ray(R). (Ray originates at the light and gets reflected on the pattern) [3]
 - b. Calculate the Diffuse Component. [2]
 - c. Calculate the Specular Component. [2]
 - d. If the radius of the light source's sphere of influence is 10 units, what will be the color of the pattern? (You need to calculate Attenuation Factor here) [3]
5. If there is a spotlight somewhere aiming at the floor, what kind of shading will you choose to view the entire floor and why? [3]

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B

Suppose you are standing on a designed floor. The floor is situated at the **XY plane**. Your eye position is **(4, 3, 6)**. There is a pattern with color— **20% red, 10% green, 70% blue**, drawn on the floor at **(1, 1, 0)**. The room has an ambient light with intensity, **$I_a = 0.3$** . Here, Ambient coefficient **$k_a = 0.3$** , Diffuse coefficient **$k_d = 0.5$** , Specular coefficient **$k_s = 0.6$** and Specular exponent **$n = 2$** . Answer the following questions based on it:

1. What is the normal vector of the plane (N)? [2]
2. Calculate the unit vector of viewing direction (V) (The vector going from pattern to Eye)? [2]
3. What color will you find in the pattern at the current setup (only ambient light)? [3]
4. Now you turned on a white light with intensity 0.7 situated at (-1, -2, 6).
 - a. Calculate the unit vector of the reflected ray(R). (Ray originates at the light and gets reflected on the pattern) [3]
 - b. Calculate the Diffuse Component. [2]
 - c. Calculate the Specular Component. [2]
 - d. If the radius of the light source's sphere of influence is 10 units, what will be the color of the pattern? (You need to calculate Attenuation Factor here) [3]
5. If there is a spotlight somewhere aiming at the floor, what kind of shading will you choose to view the entire floor and why? [3]

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C

Suppose you are standing on a designed floor. The floor is situated at the **XY plane**. Your eye position is **(4, 3, 6)**. There is a pattern with color— **60% red, 10% green, 30% blue**, drawn on the floor at **(1, 1, 0)**. The room has an ambient light with intensity, **$I_a = 0.2$** . Here, Ambient coefficient **$k_a = 0.4$** , Diffuse coefficient **$k_d = 0.6$** , Specular coefficient **$k_s = 0.5$** and Specular exponent **$n = 2$** . Answer the following questions based on it:

1. What is the normal vector of the plane (N)? [2]
2. Calculate the unit vector of viewing direction (V) (The vector going from pattern to Eye)? [2]
3. What color will you find in the pattern at the current setup (only ambient light)? [3]
4. Now you turned on a white light with intensity 0.7 situated at (-1, -2, 6).
 - a. Calculate the unit vector of the reflected ray(R). (Ray originates at the light and gets reflected on the pattern) [3]
 - b. Calculate the Diffuse Component. [2]
 - c. Calculate the Specular Component. [2]
 - d. If the radius of the light source's sphere of influence is 10 units, what will be the color of the pattern? (You need to calculate Attenuation Factor here) [3]
5. If there is a spotlight somewhere aiming at the floor, what kind of shading will you choose to view the entire floor and why? [3]